

Using HOMER software to investigate, size and apply renewable energy sources in a convention center in Sabratha, Libya.

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Abstract— The concern on dependence on fossil fuels and their environmental impact is increasing recently. This led to more research in alternative energies to reduce the reliance on fossil fuels and to protect the environment.

There are two strategies that can be applied to reduce dependence on fossil fuel. Firstly, reducing energy consumption by applying energy savings programs. And secondly, using renewable energy sources. In this paper, HOMER pro. software is applied to investigate, analyse, size, and give the optimum solution for a standalone hybrid power system to supply power to a convention center in Sabratha, Libya.

Keywords— HES, DG, HOMER, standalone power system, optimization, NPC, COE, CEQ, optimum configuration.

I. INTRODUCTION

One of the most topics debated nowadays is global warming. Burning fossil fuels produce large amount of greenhouse gas emissions, global warming is caused by increasing the quantity of these greenhouse gases into the atmosphere. Global levels of greenhouse gases have increased badly since the Industrial Revolution dawn in the 1750s [1]. The most man-made carbon dioxide (CO₂) emissions come from the burning of fossil fuels [2]. Nitrous oxide (NO_x) emissions are mostly caused by using synthetic fertilizers for agriculture, using fossil fuels for combustion and for livestock manure management [3]. The conventional fuel for power generation comes from fossil fuels, there is an increasing concern nowadays on generating electricity from fossil fuels because of their emissions, their higher prices and their exposition to depletion. The power consumption from fossil fuel resources, CO₂ emissions and low carbon energy are related based on Lotka-Voltra model [4]. There are two scenarios to apply that, either by applying energy saving programs, or by more usage of renewable energy sources to generate power. Renewable energy sources have less harm to the environment because they have either zero emissions or very few emissions compared to fossil fuels. Wind, solar, biomass,

geothermal, tides and some other clean energy sources have become better alternative clean energy source to replace or reduce generating energy from conventional sources [5, 6]. However, all these renewable energy resources are not available all the time throughout the year and have high initial costs, which make them less competitive. To overcome this challenge; relying on weather, and to have an economic, reliable and sustainable energy supply, hybrid models have been proposed by many researchers.

Hybrid combination of two or more renewable sources improve the system efficiency and performance. Hybrid renewable energy systems (HRES) consist of one renewable energy source combined with one conventional energy source or other renewable energy sources that work either in standalone mode or grid connected mode [4]. Stand-alone renewable energy systems must have enough storage capacity to manipulate power fluctuation from renewable energy sources [7]. Hybrid combination of different renewable sources make them complement each other and supply higher energy efficiency. Hybrid energy systems (HES) are more common in remote and isolated areas, moreover, they become popular in distributed generation (DG). Many HES have been installed in different countries over the last decade, which resulted in systems development and made them compete other conventional sources [8, 9]. Designing HES depend on the renewable energy sources available in the site where the hybrid system planned to be installed. In order to establish an economical and reliable HES, its design must be optimal from sizing and operation strategy viewpoint [9]. In [10 and 11] there are several studies conducted to find out the optimum sizing and the appropriate power management to meet various load demands by the users.

There are many software packages applied for HES [12, 13], hybrid optimization of multiple energy resources (HOMER) software have been extensively developed to select the optimal generation technologies and their sizes [14]. L. Olatomiwa et al.

[15] have investigated the feasibility of different configurations of power generation composed of solar panels, wind turbines and diesel generators in various geopolitical locations in Nigeria using Homer software. They concluded that the solar / wind / diesel / battery HES configuration is the optimum configuration for fuel consumption and CO₂ reduction. D. Chade et al. [16] conducted a case study of wind / hydrogen cell system in remote locations in Grimsey Island in the Arctic. They have applied HOMER software to compare the cost of electricity from fuel cost of DG and to optimize the size of the system components. They concluded that a wind turbine with hydrogen energy storage system is a feasible solution. R. Sen and S. C. Bhattacharyya [17] analyzed the optimal design and planning of an off-grid HES in a remote village in India. The study considered a combination of a small-scale hydropower, solar panels, wind turbines and bio-diesel generator. In comparing their proposed HES with conventional grid extension, they concluded that the HES is cost-effective, sustainable, techno-economically viable and environmentally friendly. HOMER software is widely used for most of the RES based systems. Thus, based on what mentioned above, HOMER software is used in this paper to study the feasibility assessment of the optimal HES in a convention center in northern west of Sabratha, Libya. The integration of renewable energy is playing a crucial rule in minimizing the CO₂ emissions, but all renewable energy sources are intermittent in nature, nonlinear and fluctuated. To tackle these issues, there should be powerful forecasting techniques as well as advanced optimization methods to improve the overall system performance and minimize the CO₂ emissions. This paper is introducing a case study of designing a hybrid model for standalone system.

II. DESIGN OF HYBRID RENEWABLE ENERGY SYSTEMS

Hybrid power generation systems are often designed to solve the power supply problem in some rural areas. The annualized cost of the systems is minimized in these hybrid systems while satisfying the required power supply.

Technology selection and system sizing:

In this design stage, the system's configuration is made, the generation technologies that will build the hybrid system will be specified. For a given hybrid energy system, it is very important to determine the type of renewable energy system to be included, the number and capacity of renewable energy units to be installed. It must be determined whether the hybrid system is stand-alone, or grid connected. Often, diesel generators, fuel cells or other back-up units are integrated in the system.

The hybrid system selection relies on the availability of the renewable resources. Based on these renewable resources, a feasibility study for various combinations is conducted using several optimization techniques to get the optimum configuration. To get an efficient system, the number and the size of the selected components need to be optimized in order to get an economical, efficient, and reliable system. There are many factors and constraints influencing the system sizing, which include greenhouse emissions, economics and reliability.

III. THE SYSTEM MODELING

The convention center is located in north west Sabratha, Libya, the altitude and the longitude of the center are 32°48.8'N and 12°22.6'E. The average temperature is °C25, and the humidity is 50%. It has Mediterranean weather. Figure (1) shows the schematic procedure to work on HOMER.

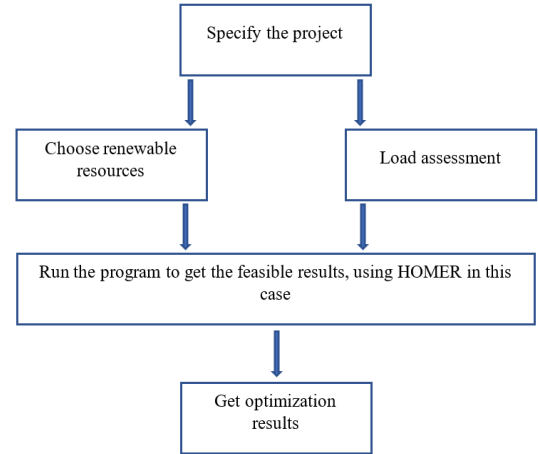


Fig. 1. Schematic procedure to run the software

IV. RESOURCE ASSESSMENT

The load profile is taken from the General Electricity Company of Libya (GECOL) record. Hybrid Optimization of Multiple Energy Resources (HOMER) software is applied to estimate the most appropriate, feasible configuration of wind, solar, diesel generators and batteries. Since renewable energy sources are intermittent in nature, back-up facilities are often used to avoid any power deficiency.

A. Wind energy source

The monthly average wind speed data was downloaded from HOMER pro software, which downloaded in return from NASA Surface meteorology and Solar Energy. The wind power equation is defined by the following equation [18]:

$$P = \frac{1}{2} \rho \cdot A \cdot C_p V^3 N_g N_b \quad (1)$$

Where, ρ is the air density in kg/m^3 ,
 A is the rotor swept area in m^2 ,
 C_p is the coefficient performance,
 V is the wind speed velocity (m/s),
 N_g is the generator efficiency,
 N_b is the gear box bearing efficiency.

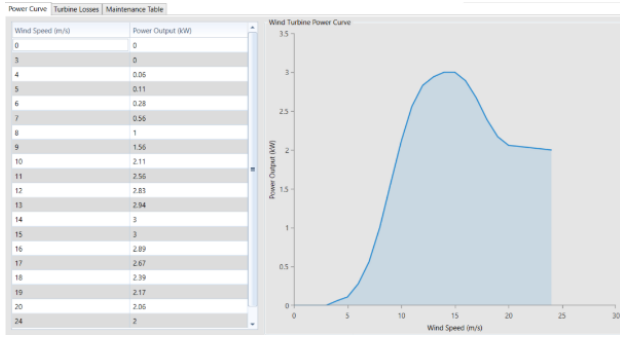


Fig. 2. Wind turbine power curve

Figure (2) shows the power curve of a wind turbine, where the electric power is depicted at various wind speeds.

B. Solar energy source

The solar data for the convention center location is taken from HOMER Pro software, which downloaded in return from NASA Surface meteorology and Solar Energy. The PV-GIS web tool uses equation (2) to calculate the annual potential of any installed PV system [19].

$$E = 365 \times P_k \times r_p \times H_{b,i} \quad (2)$$

Where, E is the annual potential for the power generated in kWh, P_k is the installed equipment peak power in kilowatts (kW), r_p is the system performance ration or derating factor and $H_{b,i}$ is the monthly or annual average of daily global radiation in wathours (wh).

V. ANALYSIS CRITERIA

Table (I) shows the energy optimized parameters of the energy resources applied. After selecting the renewable sources and running HOMER pro, the following design configurations have been established:

- Wind, diesel generator, battery and inverter model.
- Wind, battery and inverter model.
- Solar, wind, battery and inverter model.
- Solar, diesel generator, battery and inverter model.
- Solar, wind, diesel generator, battery and inverter model.

Table (I) Energy optimized parameters

Source	Optimized parameter
Wind	Wind power, cost minimization
Solar	Solar irradiation, cost minimization
Diesel Generator	CO2 emission reduction, cost minimization

These results are classified by HOMER according to NPC, COE, CEQ and some other constraints. Figure (3) shows a copy of the optimization results. The optimization results are presented in prices ascending order.

A. Net present cost (NPC)

The total net present cost in \$ as shown in figure (3) represents the present value of all the project costs during its lifetime, subtracted from that the present value of all the revenue earned over its lifetime.

PV (kW)	NPC (\$)	COE (\$)	Operating cost (\$/yr)	Initial capital (\$)
50.0	\$249,791	\$4.70	\$4,238	\$195,000
50.0	\$321,467	\$6.05	\$7,153	\$229,000
50.0	\$346,951	\$6.53	\$4,599	\$287,500
50.0	\$356,448	\$6.71	\$3,632	\$309,500
50.0	\$377,608	\$7.11	\$3,876	\$327,500

Fig. 3. The HOMER optimization results.

The costs considered here include capital costs, replacement costs, fuel costs, O&M costs, and emissions penalties. The total net present cost is calculated by HOMER as follows:

$$C_{NPC} = \frac{C_{yt}}{CRF(i_r, R_p)} \quad (3)$$

Where, C_{NPC} is the total net present cost in \$, C_{yt} is the total annualized cost in \$/year, CRF is the capital recovery factor, i_r is the interest rate in %, R_p is the project lifetime in years.

B. Cost of energy (COE)

The levelized cost of energy (COE) is defined by HOMER as the average cost per kWh of useful electrical energy produced by the hybrid system. The cost of energy (COE) is calculated by HOMER as follows:

$$COE = \frac{C_{yt}}{E_p + E_d + E_{gs}} \quad (4)$$

Where, C_{yt} is the total annualized cost in \$/year, E_p is the primary load served in kWh/year, E_d is deferrable load served in kWh/year, E_{gs} is the total grid sales.

C. Carbon emission quantity (CEQ)

CEQ is calculated by HOMER as follows:

$$CEQ = F_{fc} \times CO_2 (ef) \quad (5)$$

Where F_{fc} is fossil fuel consumption, and $CO_2 (ef)$ is the CO_2 emission factor.

VI. RESULTS AND DISCUSSION

HOMER software gives the results based on the power resources given to it and on the load demand required in the project. The simulation results will be ranked based on the total NPC. The configuration with the lowest NPC will be considered the optimal configuration. Figure (3) shows the feasible configurations ranked according to the NPC.

A. Optimization results and system analysis.

The best feasible configuration according to the optimal results given by HOMER is the model with two units of 3 kW wind turbines, one unit of 100 kW diesel generator, 50 strings of 1kWh lead acid batteries and one unit of a 100-kW system converter.

B. Electricity generation

The most feasible model according to HOMER as shown in figure (4) was the wind/ diesel generator/ battery model. In this model, 79.7% of the energy produced came from wind turbine, this is about 5304 kWh/ year. The rest 20.3% of the energy produced came from the diesel generator.



Fig. 4. Electricity generated from the wind/ generator/ battery model.

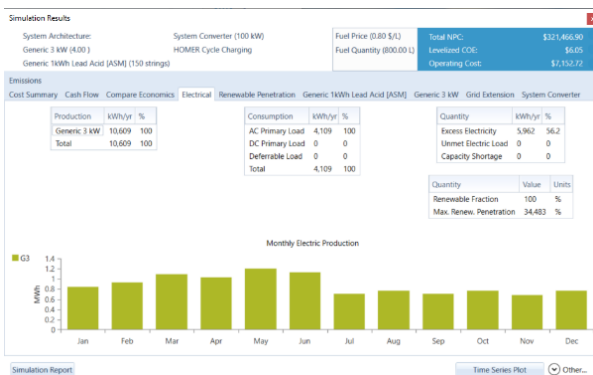


Fig. 5. Electricity generated from the wind / battery model.

The second feasible model was the wind/ battery one. 100% of the 10.609 kWh/ year came from the wind source as shown in figure (5).

The third feasible model is shown in figure (6). This model consists of solar/ wind/ battery components. 62.1% of the generated energy came from solar panels, 37.9 % came from wind generators.

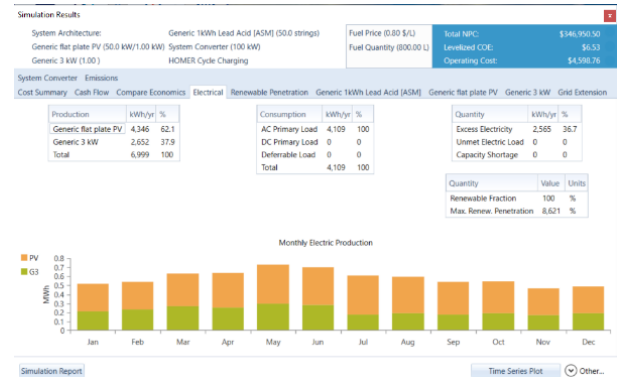


Fig. 6. Electricity generated from the solar/ wind/ battery model.

Figure (7) shows the fourth feasible model. This was the one with solar/ diesel generator/ battery. 92.5% of the energy produces was from solar, which was about 4346 kWh/year. The rest 350 kwh/year (7.45%) was from the diesel generator.

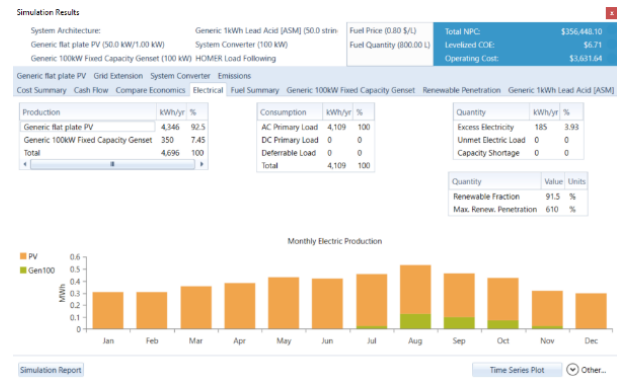


Fig. 7. Electricity generated from the solar/ generator/ battery model.

And eventually, the least feasible model shown in figure (8). This model encompasses solar/ wind/ diesel generator and battery components. 62.1% of the produced energy came from solar panels. The rest 37.9% of the energy produced came from wind turbines. The diesel generator will not be used in this model. In this paper a case study of different hybrid model is introduced to choose the optimum hybrid model with minimum CO₂ emissions.



Fig. 8. Electricity generated from the solar/ wind/ generator/ battery model.

C. Emission reduction

Even though the wind/ generator/ battery option is the cheapest one as shown in figure (3), it has the worse emission value as shown in table (II). On the other hand, the solar/ wind/ generator battery is the expensive one, but it has zero emissions, because the generator was not used in this case, but kept stand by for contingency cases. Trade-off must be taken between these options to have continuous power supply and low emissions value.

Table (II) Emissions value in kg/year

	CO ₂ (kg/yr)	CO (kg/yr)	UH (kg/yr)	PM (kg/yr)	SO ₂ (kg/yr)	NO _x (kg/yr)
W, G, B	1289	8.77	0.355	0.0351	3.16	0.701
W, B	0	0	0	0	0	0
S, W, B	0	0	0	0	0	0
S, G, B	334	2.27	0.092	0.091	0.819	0.182
S, W, G, B	0	0	0	0	0	0

Where:

UB: Unburned hydrocarbons

PM: Particulate matter

(kg/yr): kilogram / year

W: Wind, G: Diesel generator, B: Battery, S: Solar

D. Cost benefits

The total NPC, Levelized COE and the operation cost of each model are shown in figures (4) to (8) on the top right of each figure. The wind/ generator/ battery model has the least total NPC. Table (III) shows detailed prices of each model.

Table (III) detailed prices of the optimum configurations.

The model	Total NPC in (\$)	Levelized COE in (\$)	Operating cost in (\$)
W, G, B	249,790.80	4.7	4,238.31
W, B	321,466.90	6.05	7,152.72
S, W, B	346,950.5	6.53	4,598.76
S, G, B	356,448.10	6.71	3,631.64
S, W, G, B	377,607.80	7.11	3,876.06

Figures (9) and (10) show one-week time series renewable load served data for three different models. Where the dark color represents the total electrical load served, the green color represents the renewable penetration and the yellow color represents the total renewable power output.

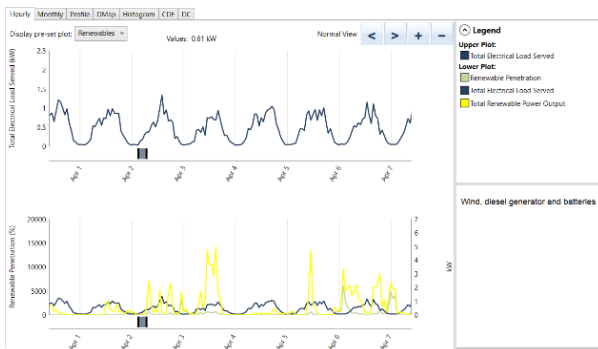


Fig. 9. One-week time series renewable load served data.

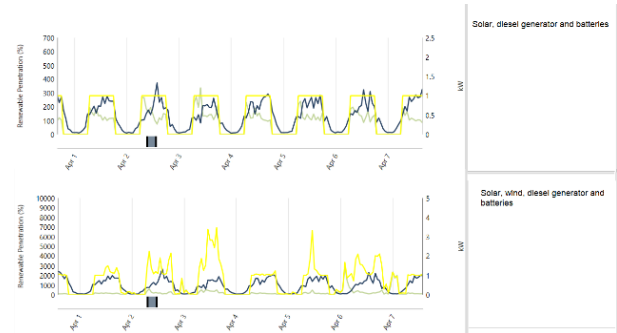


Fig. 10. One-week time series renewable load served data for another two models.

VII. CONCLUSION

Electric power is one of the most important things in our daily life. Generating electricity from the conventional resources has adversely impact in the environment. Replacing either all these conventional sources or part of them mitigate their impact by reducing Greenhouse emissions produced from them. In this paper, HOMER pro software is used to apply some renewable energy sources on a convention center in Sabratha, Libya. The results show that the electric power can be generated from clean energy sources based on different hybrid models. The simulation results were ranked according to their total NPC from the least to the highest price, where the least one considered the optimal configuration. Care must be taken when considering the optimal configuration, because the cheapest option has the worst pollutant emissions. Trade-off must be taken between the projected models to get continuous power supply with less emissions. Optimal operation of power systems is depending on a highly accurate forecasting models as renewable are fluctuated, intermittent and nonlinear sources of energy. The future work will be focused on creating an online forecasting model for different energy resources and connect it to some optimizing techniques to create 24 hours ahead highly accurate hybrid model with minimum cost and CO₂ emissions.

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